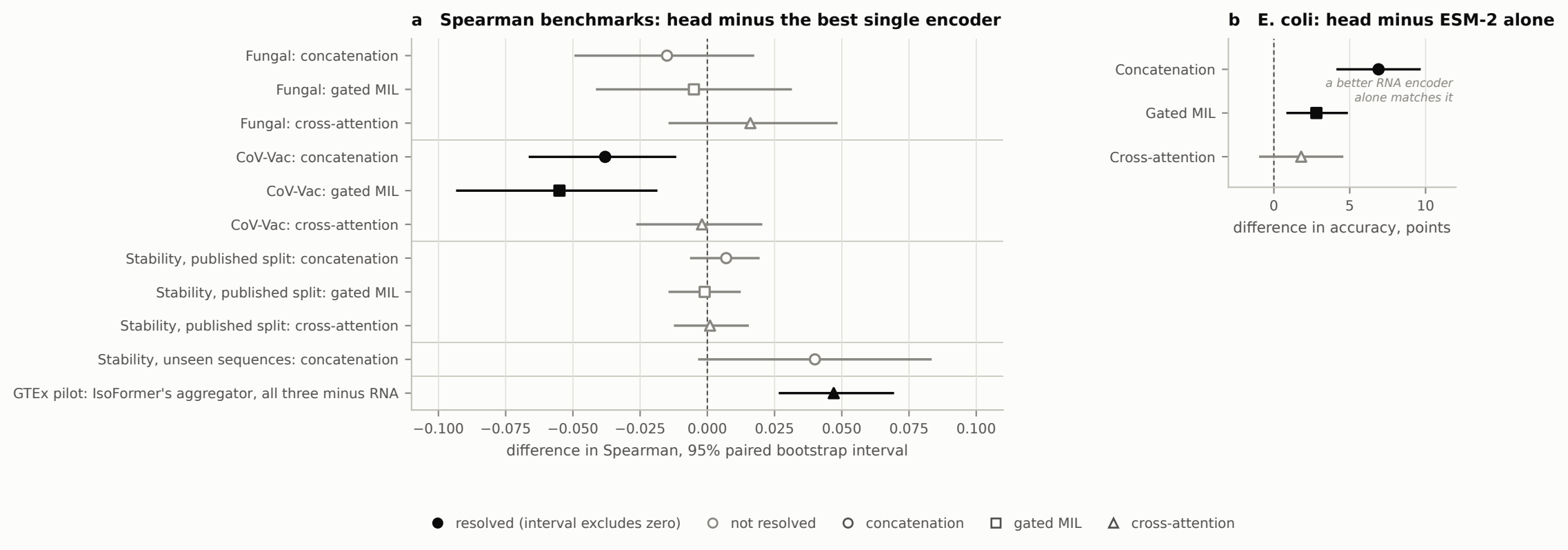
a Spearman benchmarks: head minus the best single encoder
Fungal: concatenation
Fungal: gated MIL
Fungal: cross-attention
CoV-Vac: concatenation
CoV-Vac: gated MIL
CoV-Vac: cross-attention
Stability, published split: concatenation
Stability, published split: gated MIL
Stability, published split: cross-attention
Stability, unseen sequences: concatenation
GTEx pilot: IsoFormer's aggregator, all three minus RNA
−0.100 −0.075 −0.050 −0.025 0.000 0.025 0.050 0.075 0.100
difference in Spearman, 95% paired bootstrap interval
b E. coli: head minus ESM-2 alone
Concatenation
Gated MIL
Cross-attention
a better RNA encoder alone matches it
0 5 10
difference in accuracy, points
resolved (interval excludes zero)
not resolved
concatenation
gated MIL
cross-attention